

Grant Keefe

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Education

Queen's University, BAS in Mechatronics and Robotics Engineering Sept 2021 - May 2026

- Principal's Scholar, Dean's Scholar: Upper year GPA 4.1/4.3
- **Coursework:** Microprocessor Systems, Computer Networks, Mobile Robotics, Integrated Circuits

Professional Experience

R&D Contractor, Dominion Dynamics – Kingston, ON Sept 2025 – May 2026

- Built a persistent Arctic sensing node designed to serve both military and civilian purposes with a budget of \$100,000, the company is currently pursuing a patent for the design.

Research Assistant, Ingenuity Labs Research Institute – Kingston, ON Sept 2025 – May 2026

- Wrote my thesis on the use of novel sensor modalities for anomaly detection in Maritime environments, Queen's university is pursuing a patent for this design.

SAR Robotics Researcher, Ingenuity Labs Research Institute – Kingston, ON Sept 2024 – Sept 2025

- Lead author of grant application and receiving \$515,000 in funding and an additional \$135,000 in institutional support to investigate fixed-wing UAV applications for maritime search and rescue.
- Designed the mechanical, electrical, and sensor processing for a waterproof fixed-wing UAV, conducted flight testing, controller tuning and validated long-range communications links.

Electrical Engineering Co-op, GeoSpectrum Technologies Inc. – Halifax, NS May 2024 – Sept 2024

- Test engineer in R&D department designing and building methods for testing and troubleshooting sub-systems.

Teacher's Assistant, Queen's University – Kingston, ON Dec 2023 – May 2024

- Helped students successfully integrate ROS2 into their robot designs and take their first steps towards full autonomy.

Junior Automation Engineer, Enginuity Inc. – Halifax, NS May 2023 – Sept 2023

- Designed and tested control panel for water mixing and dosing system. Design work done in SolidWorks and AutoCAD Electrical.

Head Sailing Instructor, Royal Nova Scotia Yacht Squadron – Halifax, NS May 2022 – Sept 2022

- Oversaw daily operations and a team of instructors, developing leadership, communication, and organization. I was responsible for the water safety of students, and aware of challenges in rescue in marine environment.

Projects

Thrust Vectored Rocket Design Project Page

- Designed a full preliminary rocket prototype featuring thrust-vectoring gimbal, four-servo canard box with encoder feedback, and a deployable nose-drogue for passive descent stabilization.

Ball Gimbal Project Page

- Designed and built a 2-DOF ball gimbal for a nose mounted FPV and depth camera. The gimbal was stable and controllable during flight testing.

Unmanned Aerial System Competition Project Page

- I designed and assembled 6S5P lithium-ion battery packs, created the team's first power distribution board, conducted the iron-bird testing, did the full wiring and subsystem integration, and contributed to structural assembly and flight-testing.
- The aircraft earned 1st place in Prototype Realism and 2nd place in Pitch Presentation at the 2024 AEAC competition.

Micro Air Vehicle Competition Project Page

- Designed and built the competition drone. I collaborated with the team to develop and validate the ROS2 planning and control across Gazebo simulation and real-world testing.
- The team placed 2nd among undergraduate teams at the competition.

Extracurricular

Defense Tech Hackathon, First Place Team - \$10,000 Nov 2025

- Processed labeled drone audio data with a log-mel spectrogram sliding window, then trained YOLO11n-cls arch. on spectrograms achieving a validation and test accuracy of 95%

Captain, Queen's Aerospace Design Team

May 2023 – Sept 2024

- Responsible for managing a team of 60 students across seven sub-teams with a total operating budget of \$40,000.

Software/Electrical Lead, Queen's Aerospace Design Team

Sept 2022 – May 2023

- As a Software Lead utilized C and data structures to develop a route optimization algorithm and implemented as an offboard controller sending serialized commands to flight controller.

Lacrosse Team, Queen's University

Sept 2022 – May 2026

- 15-20 hour a week commitment

Technical Skills

Programming: Git, MATLAB, Simulink, ROS2, Gazebo, C, C++, Python, Arduino, PX4, Ardupilot

Mechanical Design: SolidWorks + Simulation, FDM 3D Printing, SLA Printing, Laser Cutting

Electrical Design: Ki-CAD, AutoCAD Electrical, STM32 cube mx, Surface mount device (SMD) soldering, Signal probing and analysis